

Recommendation Systems

Jaccard Similarity

- **Jaccard Similarity** = (number of observations in both sets) / (number in either set)
- mathematically,

$$J(A, B) = |A \cap B| / |A \cup B|$$

Jaccard Similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	4			5	1		
<i>B</i>	5	5	4				
<i>C</i>				2	4	5	
<i>D</i>		3					3

- $\text{sim}(A,B) = |r_A \cap r_B| / |r_A \cup r_B|$

- $\text{sim}(A,B) = 1/5; \text{sim}(A,C) = 2/4$

Cosine Similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4	0	0	5	1	0	0
B	5	5	4	0	0	0	0
C				2	4	5	
D		3					3

- $\text{sim}(A,B) = \cos(r_A, r_B)$
- $\text{sim}(A,B) = 0.38$, $\text{sim}(A,C) = 0.32$
 - $\text{sim}(A,B) < \text{sim}(A,C)$, but not by much

Pearson Correlation (Centered Cosine)

- Normalize rating by subtracting the mean

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	2/3			5/3	-7/3		
B	1/3	1/3	-2/3				
C				-5/3	1/3	4/3	
D		0					0

- $\text{sim}(A,B) = \cos(r_A, r_B) = 0.09$; $\text{sim}(A,C) = -0.56$
 - $\text{sim}(A,B) > \text{sim}(A,C)$
- Captures intuition better
 - Missing ratings treated as “average”
 - Handles “tough raters” and “easy raters”

Text Similarity

- Text similarity refers to measuring how similar two pieces of text are, based on their content, structure, or semantic meaning.
- **TF-IDF:**
- **Term Frequency (TF):** Measures how often a word appears in a document.
- **Inverse Document Frequency (IDF):** Measures how unique or rare a word is across a collection of documents.
- The combination helps in weighting the words and improving the cosine similarity measure by reducing the impact of common words.

Cosine Similarity

- **Text Preprocessing:**

- **Tokenization:** Split the text into individual words or tokens.
- **Lowercasing:** Convert all text to lowercase to ensure uniformity.
- **Stopword Removal:** Remove common words (like "the", "and", "is") that do not contribute much to the meaning.
- **Stemming/Lemmatization:** Reduce words to their base or root form.

- **Vector Representation:**

- **Term Frequency (TF):** Count the frequency of each word in the document.
- **Inverse Document Frequency (IDF):** Calculate the importance of a word based on its rarity across multiple documents.
- **TF-IDF:** Combine TF and IDF to get a weighted score that reflects the importance of a word in a document relative to a collection of documents.

Cosine Similarity

- **Construct Vectors:**

- Represent each document as a vector of TF-IDF scores, where each dimension corresponds to a unique word in the corpus.

- **Calculate Cosine Similarity:**

- Use the cosine similarity formula to compute the similarity between two document vectors.

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}},$$

Cosine Similarity (Example)

- **Text A:** "I love machine learning and data science."
- **Text B:** "Data science and machine learning are fascinating."
- **Preprocessing:**
After tokenization and stopwords removal:
 - TextA: ["love", "machine", "learning", "data", "science"]
 - TextB: ["data", "science", "machine", "learning", "fascinating"]
- **Vector Representation**
Create a vocabulary:
["love", "machine", "learning", "data", "science", "fascinating"]

Calculate TF-IDF for each word in both texts.

Cosine Similarity (Example)

- TF-IDF Vectors:
 - TextA: [1, 1, 1, 1, 1, 0]
 - TextB: [0, 1, 1, 1, 1, 1]

$$\begin{aligned}\text{Cosine Similarity} &= \frac{(1 \times 0) + (1 \times 1) + (1 \times 1) + (1 \times 1) + (1 \times 1) + (0 \times 1)}{\sqrt{1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 0^2} \times \sqrt{0^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2}} \\ &= \frac{4}{\sqrt{5} \times \sqrt{5}} = \frac{4}{5} = 0.8\end{aligned}$$

Types of Recommendation Systems

- Content-based filtering
- Collaborative filtering
 - User-user collaborative filtering
 - Item-item collaborative filtering
- Hybrid systems
- Knowledge-based recommendations
- Context-aware recommendation systems

Collaborative Filtering

- Collaborative filtering is a popular technique in recommendation systems that predicts a user's preferences by leveraging other users' preferences.
- There are two main types of collaborative filtering:
 - **user-user collaborative filtering**, and
 - **item-item collaborative filtering**.

User-Based Collaborative Filtering:

- This approach predicts a user's rating for an item based on ratings given by other users who have similar preference

Step1: User Similarity Computation

- Calculate the similarity between users using metrics like **Jaccard similarity, cosine similarity, or Pearson correlation**.

Step 2: Neighborhood Formation:

- Identify a group of users (neighbors) who are most similar to the target user.

Step 3: Prediction

- Predict the rating for an item by averaging the ratings given by similar users, possibly weighted by their similarity scores.

Item-Based Collaborative Filtering:

- This method predicts a user's rating for an item based on the user's previous ratings of similar items.
- **Example:** If a user likes movie A and movie B is similar to movie A, the system can recommend movie B to the user.
- **Steps:**
 - 1. Item Similarity Computation
 - **2. Neighborhood Formation:**
 - 3. Prediction

Example

User-Item Rating Matrix:

Movie	User 1	User 2	User 3	User 4
Movie A	5	4	4	?
Movie B	3	5	?	3
Movie C	4	?	5	4
Movie D	?	3	4	5
Movie E	2	2	?	3

Our goal is to predict User 4's rating for **Movie A**.

Model-Based Collaborative Filtering:

- Utilizes machine learning models to learn user-item interactions and make predictions.
- Techniques:
 - **Matrix Factorization (e.g., SVD, ALS):** Decomposes the user-item matrix into latent factors representing users and items.
 - **Deep Learning:** Neural networks to capture complex user-item interactions.

Example: Netflix's recommendation system uses matrix factorization to predict user preferences for movies.

Cold Start Problem

- The cold start problem occurs when a recommendation system struggles to make accurate suggestions due to insufficient data about **new users, new items**, or both.
- Types of Cold Start Problems
 - New User
 - New Item
 - New System

Cold Start Problem- Common Solutions

- For New User Cold Start
 - Onboarding questionnaires/preference elicitation
 - Demographic-based recommendations
 - Popular items / trending courses
 - Hybrid systems
- For New Item Cold Start
 - Content-based metadata
 - Expert curation
- For New System Cold Start
 - Seed data
 - Transfer learning / Cross-domain recommendation
- **Advanced Techniques**